

Pharmacology: Allergens and Allergic Rhinitis

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Objectives

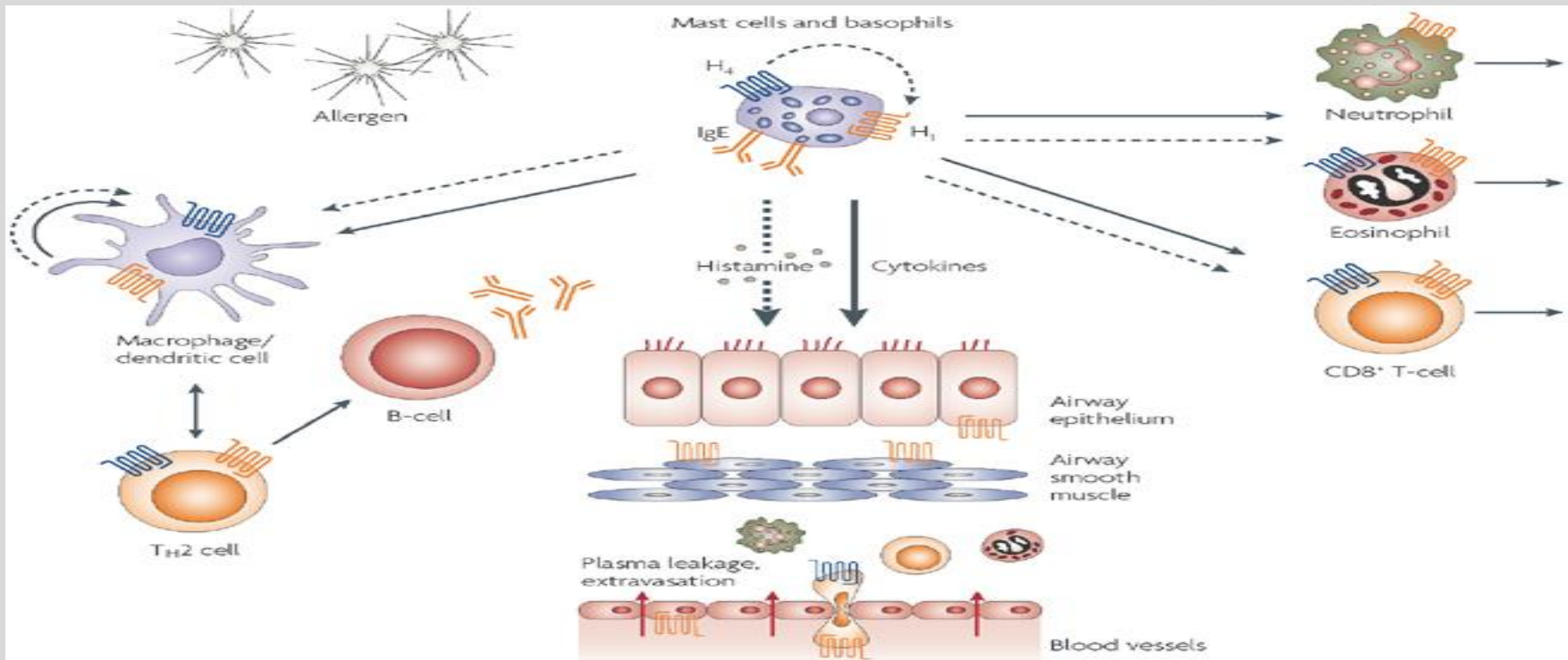
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- **Describe the role of histamine in allergic reactions**
- **Examine the physiology of the H₁ receptor**
- **Distinguish between first-and second-generation antihistamines. Include adverse effects.**
- **Describe other drugs used in the management of allergic reactions.**

Histamine

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Adverse effects of histamine release

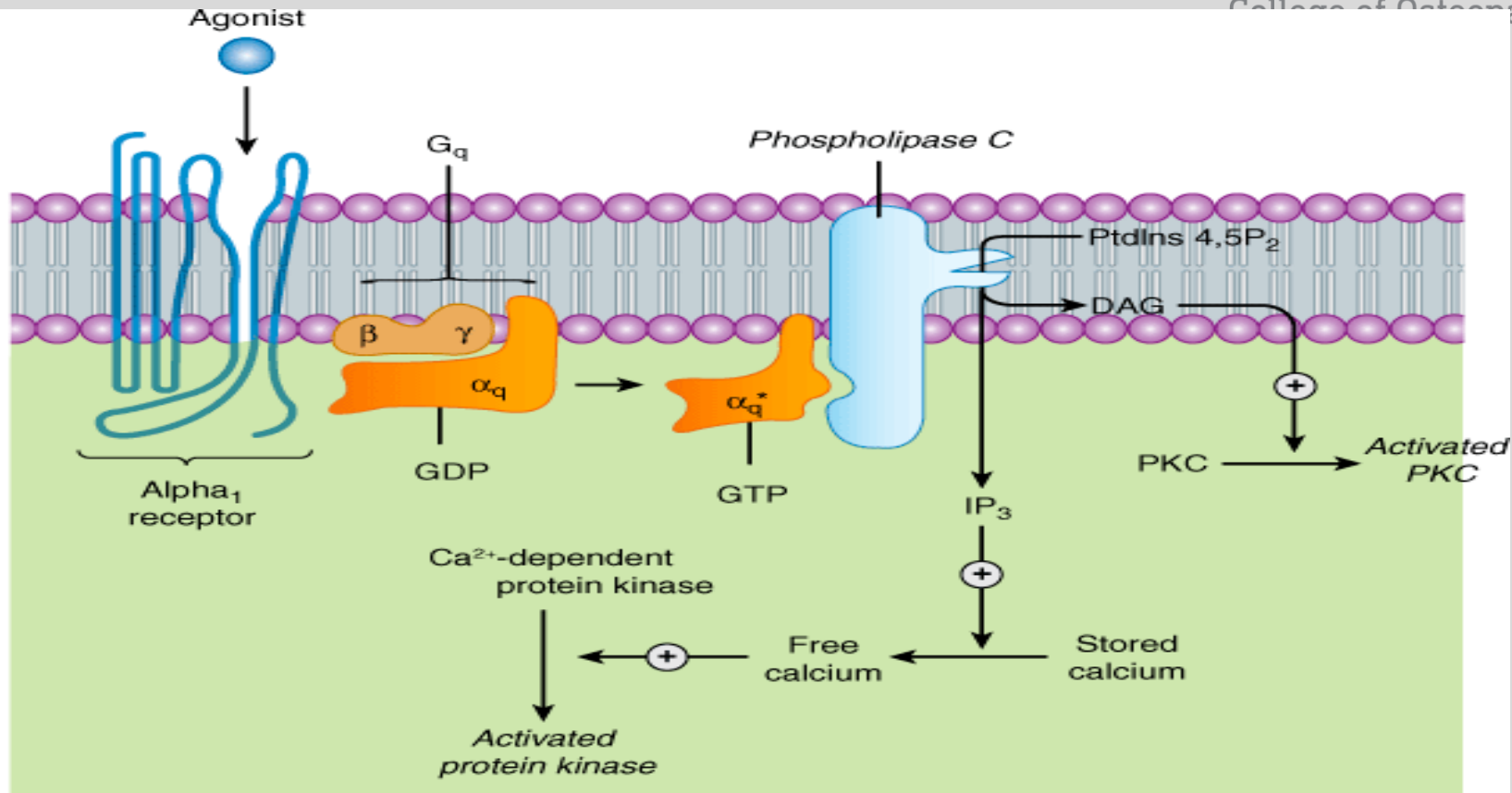
- Itching, Urticaria
- Flushing
- Hypotension
- Tachycardia
- Bronchospasm
- Angioedema
- Wakefulness
- Increased acidity (Gastric acid secretion)



Alpha₁, M₃, H₁

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Antihistamines

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- **1st Generation: Drowsy**
- **2nd Generation: Non-Drowsy**

H₁ Receptor Blockers

H₁ Receptor Blockers (First Generation)

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DRUG	MOA	USEFULNESS	ADVERSE EFFECTS
Dimenhydrinate (Dramamine®)	H₁ receptor blocker Anticholinergic (M)	(Anticholinergic) Antimotion Antiemetic (Antihistaminic) Seasonal allergy Hay Fever Dermatitis	Dryness Strong sedation (advantage at bedtime) Blurred vision (glaucoma warning) Weight gain Urinary retention (BPH) Constipation, increased AV node action
Diphenhydramine (Benadryl®) Chlorpheniramine			
Promethazine (Phenergan®)	As above, plus Alpha-1 and D₂ Block	As above	As above plus Orthostatic hypotension Pseudoparkinsonism
Cyproheptadine	As above, plus antiserotonergic 5-HT₂	As above Serotonin Syndrome	As above, but less Alpha- 1 block

H₁ Receptor Blockers (Second Generation)

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DRUG	MOA	USEFULNESS	ADVERSE EFFECTS
Fexofenadine (Allegra®)	Competitive H₁ antagonists No Anticholinergic action	Non-Sedating. A major advantage for daytime treatment of allergies, urticaria, hay fever, etc.	No antiemetic effect. No benefit against motion sickness. No CNS distribution, less sedation. Not lipophilic, no BBB cross
Loratadine (Claritin®) Desloratadine (Clarinex®)			
Cetirizine (Zyrtec®) Levocetirizine (Xyzal®)			

Other Modifiers of Histamine

Action and Release

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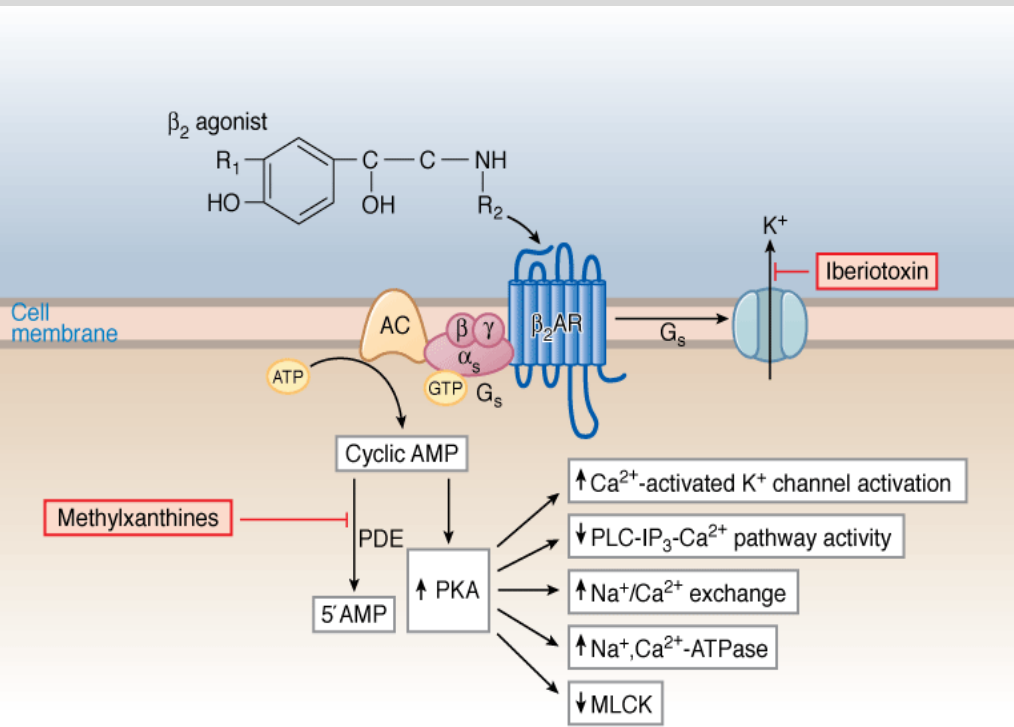
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DRUG	MOA	USEFULNESS	ADVERSE EFFECTS
Epinephrine Ephedrine	Beta-1, Beta-2, alpha-1, alpha 2 agonist	Reverse bronchoconstriction Rescue from anaphylaxis	Skeletal muscle tremor; cardiac palpitations. Epinephrine may also cause CNS excitement and hypertension, arrhythmia.
Selective B₂ Agonists Albuterol (SABA) Salmeterol (LABA)	Beta-2 agonists	Asthma/COPD	
Cromolyn sodium Nedocromil	Stabilize mast cells against antigen IgE- sensitized release of histamine	“Prophylactic”	Very safe as inhaled and not absorbed.
Lodoxamide Ketotifen (ophthalmic)			

Beta₂ Receptor (Gs)

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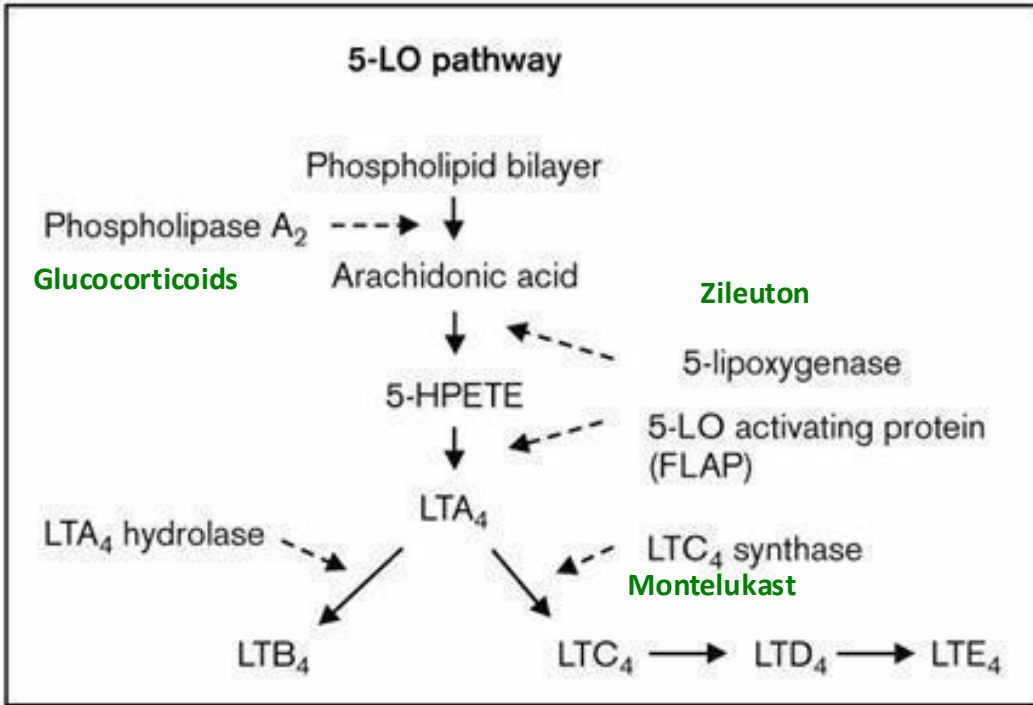
Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*:
www.accessmedicine.com

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Eicosanoids

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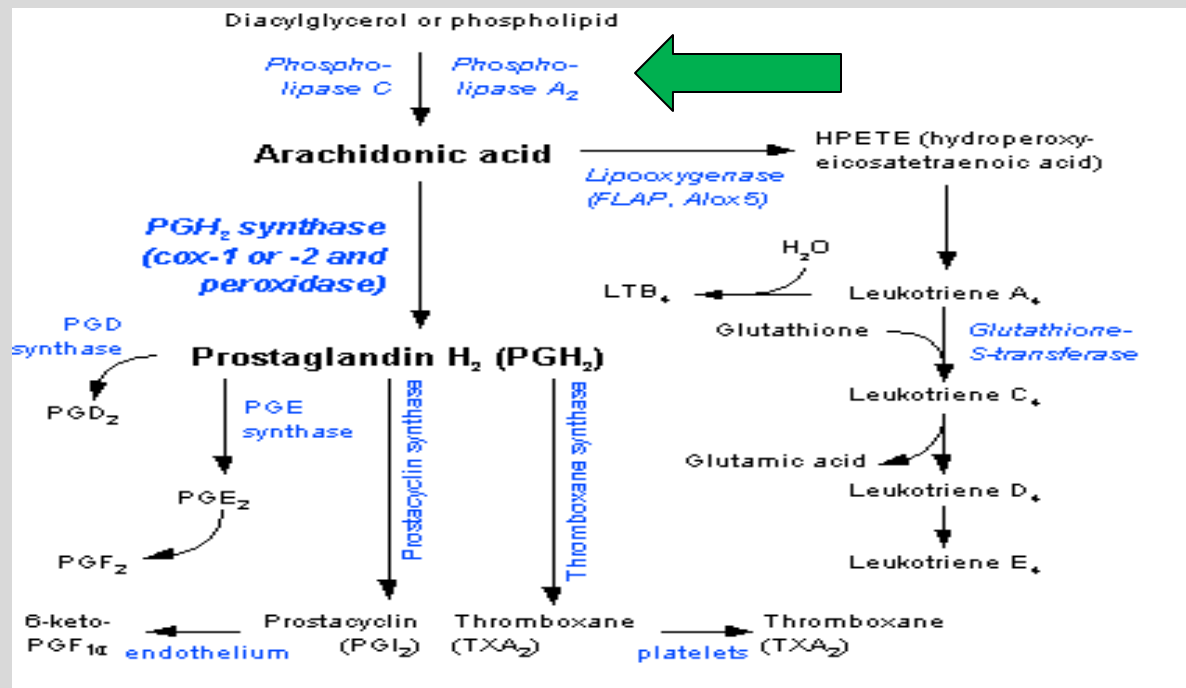
Glucocorticoids

- Stimulate synthesis of lipocortin

- Lipocortin inhibits Phospholipase A₂
 - Blocks starting point for prostaglandin, leukotriene synthesis

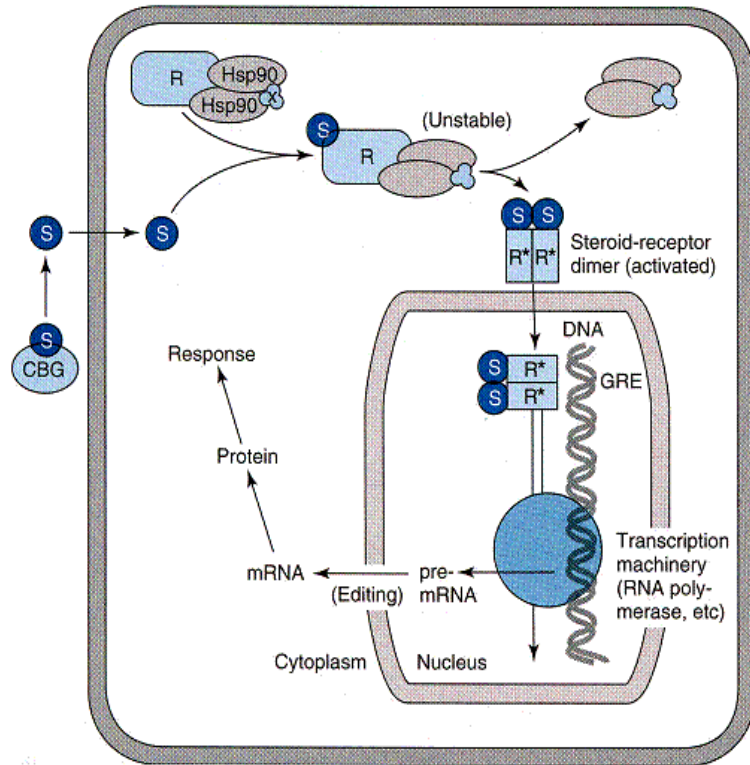
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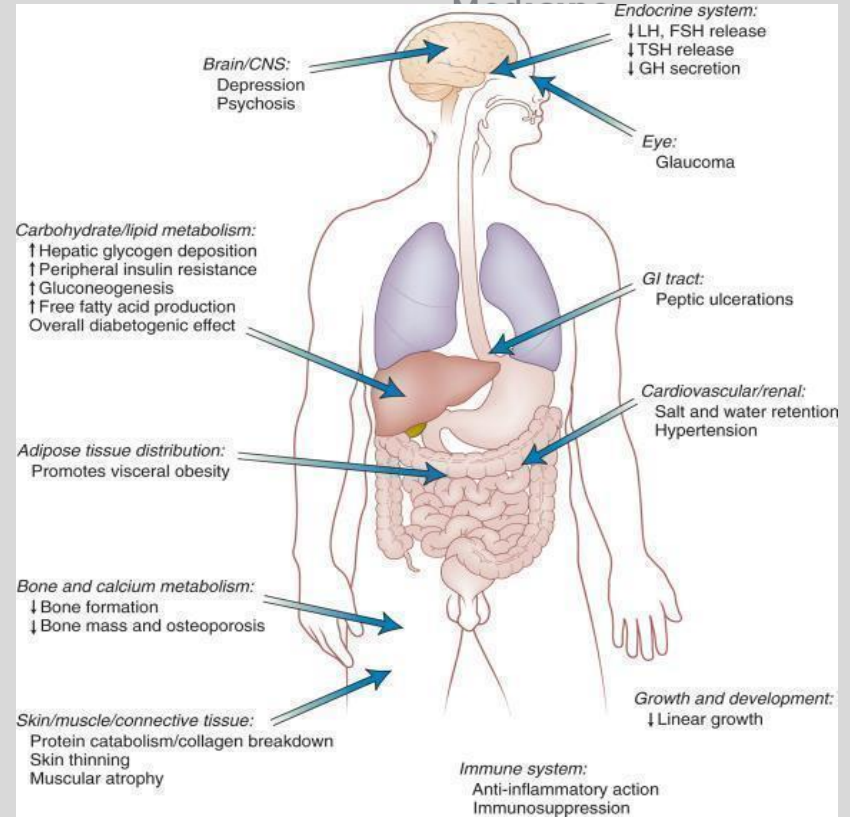
Glucocorticoids

HPA suppression: taper dose slowly



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Question

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A 51-year-old airplane pilot presented for his annual physical examination and complained of excessive itchy, watery eyes and sneezing associated with seasonal allergies. Which of the following is the most appropriate drug for this patient?

- A. Cyproheptadine**
- B. Diphenhydramine**
- C. Epinephrine**
- D. Fexofenadine**
- E. Prednisone**

Question

A 43-year-old male with a history of seasonal allergies develops wheezing, chest tightness, and bronchoconstriction soon after starting aspirin to prevent cardiovascular disease. He is administered montelukast. Which of the following describes the mechanism of the drug?

- A. 5HT_{2A} antagonist**
- B. LTD₄ antagonist**
- C. M₃ antagonist**
- D. Inhibitor of phospholipase A₂**
- E. Inhibitor of 5-lipoxygenase**

Question

A 17-year-old female developed itching soon after consuming shrimp. Minutes later, she became anxious that her chest is tight and starts gasping for air. She was administered a drug and within minutes, the episode resolves. Which of the following describes the receptor which reversed her difficulty breathing?

- A. Alpha₁ agonist**
- B. Beta₁ antagonist**
- C. Beta₂ agonist**
- D. Mast cell stabilizer**
- E. M₃ agonist**

Summary Slide

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- **Distinguish between first-and second-generation antihistamines. Include adverse effects.**
- **Describe other drugs used in the management of allergic reactions.**

Course Materials

- **Basic & Clinical Pharmacology 16e, 2024; Todd W. Vanderah; Chapter 16: Histamine, Serotonin, & the Ergot Alkaloids; Chapter 18: The Eicosanoids: Prostaglandins, Thromboxanes, Leukotrienes, & Related Compounds; Chapter 20: Drugs Used in Asthma & Chronic Obstructive Pulmonary Disease**
- **Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 34: Histamine, Bradykinin, and Their Antagonists**
- **Allergens and Allergic Rhinitis Summary and Practice Questions (Maria A. Pino, Ph.D.)**

Lecture Feedback Form:

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